

Cloud Computing Practical Examination

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Question 1

The screenshot displays a MobaXterm terminal window with two sessions. The top session, titled '98.81.181.2 (ec2-user)', shows the installation of Docker on an Amazon Linux 2 instance. It lists the running scriptlets, packages being installed, and verification steps for various components like container-selinux, docker, containerd, iptables, libgroup, libnetfilter, libnetlink, libnftnl, and pigz. The installation is confirmed as complete, and the Docker service is started and enabled. The bottom session, titled 'ip-172-31-17-194.ec2.internal', shows a transaction check for the same packages, which is successful. The terminal window includes a sidebar with file explorer and session management options, and a status bar at the bottom showing system information and a warning for the unregistered version of MobaXterm.

```
Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch 10/11
Running scriptlet: docker-25.0.16-1.amzn2023.0.2.x86_64 11/11
Installing : docker-25.0.16-1.amzn2023.0.2.x86_64 11/11
Running scriptlet: docker-25.0.16-1.amzn2023.0.2.x86_64 11/11
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket -> /usr/lib/systemd/system/docker.socket.

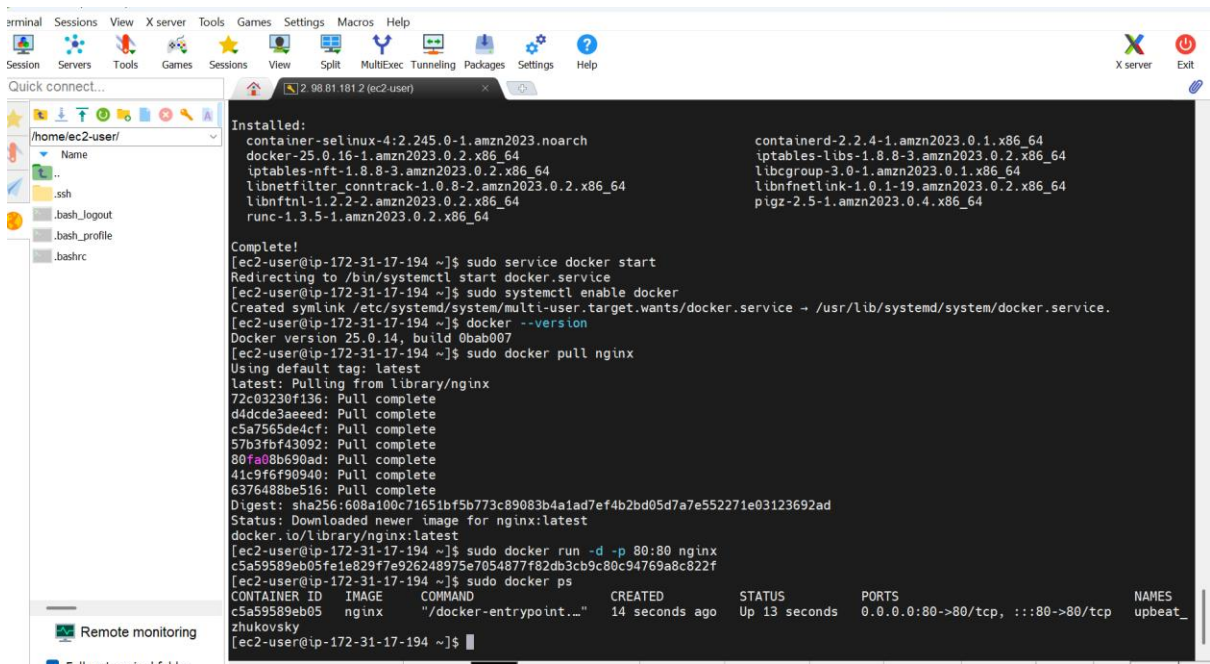
Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch 11/11
Running scriptlet: docker-25.0.16-1.amzn2023.0.2.x86_64 11/11
Verifying : container-selinux-4:2.245.0-1.amzn2023.noarch 1/11
Verifying : containerd-2.2.4-1.amzn2023.0.1.x86_64 2/11
Verifying : docker-25.0.16-1.amzn2023.0.2.x86_64 3/11
Verifying : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64 4/11
Verifying : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 5/11
Verifying : libgroup-3.0-1.amzn2023.0.1.x86_64 6/11
Verifying : libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 7/11
Verifying : libnetlink-1.0.1-19.amzn2023.0.2.x86_64 8/11
Verifying : libnftnl-1.2.2-2.amzn2023.0.2.x86_64 9/11
Verifying : pigz-2.5-1.amzn2023.0.4.x86_64 10/11
Verifying : runc-1.3.5-1.amzn2023.0.2.x86_64 11/11

Installed:
container-selinux-4:2.245.0-1.amzn2023.noarch containerd-2.2.4-1.amzn2023.0.1.x86_64
docker-25.0.16-1.amzn2023.0.2.x86_64 iptables-libs-1.8.8-3.amzn2023.0.2.x86_64
iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 libgroup-3.0-1.amzn2023.0.1.x86_64
libnetfilter_conntrack-1.0.8-2.amzn2023.0.2.x86_64 libnetlink-1.0.1-19.amzn2023.0.2.x86_64
libnftnl-1.2.2-2.amzn2023.0.2.x86_64 libnftnl-1.2.2-2.amzn2023.0.2.x86_64
runc-1.3.5-1.amzn2023.0.2.x86_64 pigz-2.5-1.amzn2023.0.4.x86_64

Complete!
[ec2-user@ip-172-31-17-194 ~]$ sudo service docker start
Redirecting to /bin/systemctl start docker.service
[ec2-user@ip-172-31-17-194 ~]$ sudo systemctl enable docker
Created symlink /etc/systemd/system/multi-user.target.wants/docker.service -> /usr/lib/systemd/system/docker.service.
[ec2-user@ip-172-31-17-194 ~]$ docker --version
Docker version 25.0.14, build 0bab007
[ec2-user@ip-172-31-17-194 ~]$
```

```
Total
Running transaction check
Transaction check succeeded.
Running transaction test
Transaction test succeeded.
Running transaction
Preparing : runc-1.3.5-1.amzn2023.0.2.x86_64 1/1
Installing : containerd-2.2.4-1.amzn2023.0.1.x86_64 1/11
Installing : containerd-2.2.4-1.amzn2023.0.1.x86_64 2/11
Running scriptlet: containerd-2.2.4-1.amzn2023.0.1.x86_64 3/11
Installing : pigz-2.5-1.amzn2023.0.4.x86_64 3/11
Installing : libnftnl-1.2.2-2.amzn2023.0.2.x86_64 4/11
Installing : libnetlink-1.0.1-19.amzn2023.0.2.x86_64 5/11
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Installing : iptables-libs-1.8.8-3.amzn2023.0.2.x86_64 7/11
Installing : iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 8/11
Running scriptlet: iptables-nft-1.8.8-3.amzn2023.0.2.x86_64 8/11
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Running scriptlet: docker-25.0.16-1.amzn2023.0.2.x86_64 10/11
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Running scriptlet: docker-25.0.16-1.amzn2023.0.2.x86_64 11/11
Created symlink /etc/systemd/system/sockets.target.wants/docker.socket -> /usr/lib/systemd/system/docker.socket.

Running scriptlet: container-selinux-4:2.245.0-1.amzn2023.noarch 11/11
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Verifying : container-selinux-4:2.245.0-1.amzn2023.noarch 1/11
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Verifying : libnetlink-1.0.1-19.amzn2023.0.2.x86_64 8/11
```



q

```
er@ip-172-31-17-194 ~$ docker --version
er version 24.0.5, build 17379ea
```

```
er@ip-172-31-17-194 ~$ sudo docker images
```

REPOSITORY	IMAGE	COMMAND	CREATED	STATUS
	nginx	-g 'daemon # 'lxcnottal	15 seconds ago	

```
er@ip-172-31-17-194 ~$ sudo docker ps
```

IMAGE	COMMAND	CREATED	PORTS
nginx	-g 'daemon off; nginx -tdecdey;	15 seconds ago	0.0.0.0:80->80/tcp, :::80->80/tcp

```
ec2-user@ip-172-31-17-194_sfor images
```

pninx	Tag	image ID	Size
nginx	latest	2f116eaf22d627	312MB

```
ec2-user@ip-172-31-17-194_sfor docker p
```

IMAGE ID	CREATED	STATUS	PORTS	NAMES
ea1d43cda5c51g5s	15 seconds ago	Up 14 seconds	0.0.0.0:80~80/tcp->80:80, :::80->80/tcp	focus-shtern

Welcome to nginx!

If you see this page, the nginx web server is successfully installed and working,

Further configuration is required.

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Question 2

```
ec2-user@ip-172-31-17-194 ~$ sudo  
docker volume create studentdata  
studentdata
```

```
ec2-user@ip-172-31-17-194 ~$  
sudo docker run -it -v studentdata:/data ubuntu  
root@b8ec8f4823fe:/% ls /data  
info.txt
```

```
root@b8ec8f4823fe:/#  
echo docker ps  
root@b8ec8f4823fe:/# echo "Rehan, RollNo123"  
█
```

```
root@b8ec8f4823fe:/#  
cat /data/info.txt  
Rehan, RollNo123
```

Question 3

*Dockerfile x

1. FROM ubuntu

2. CMD echo "Cloud Computing Practical Exam Completed Successfully"

ec2-user@ip-172-31-17-194 ~\$ sudo docker image

REPOSITORY	TAG	IMAGE ID	CREATED	SIZE
myexamimage	latest	23f8842a1cd6	About a minute ago	2.048kB
nginx	latest	cf16eaf2d627	4 days ago	1.024kB

ec2-user@ip-172-31-17-194 ~\$ sudo docker images

ec2-user@ip-172-31-17-194 ~\$ sudo docker build -t myexamimage .

Sending build context to Docker daemon 2.048kB

Step 1/2 : FROM ubuntu

--> aed43bb4c8be

Step 2/2 : CMD echoCloud Computing Practical Exam Completed Successfully

--> Using cache

---> 23f8842a1cd6

Successfully built 23f8842a1cd6

Successfully tagged myexamimage:latest

ec2-user@ip-172-31-17-194 ~\$ sudo docker run myexamimage

ec2-user@ip-172-31-17-194 ~\$ sudo docker run myexamimage

Cloud Computing Practical Exam Completed Successfully